

A Unified Ensemble Learning Framework for Software Effort Estimation Across Multiple Schemas

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The Problem: Software Effort Estimation (SEE)

Why It Matters:

- Chaos Report (Standish Group): only **29%** of projects succeed
- **52%** challenged (delayed/over-budget), **19%** fail completely
- Root cause: **inaccurate effort estimation**

Current Methods & Limitations:

- **Parametric** (COCOMO II): transparent but rigid on heterogeneous data
- **ML** (RF, XGBoost): flexible but *lacks a reproducible, fair evaluation framework*

Three Sizing Schemas in Practice

- **LOC** (Lines of Code): 2,765 projects — 90.5%
- **FP** (Function Points): 158 projects — 5.2%
- **UCP** (Use Case Points): 131 projects — 4.3%

Core Problem

LOC, FP, and UCP are studied in **isolation**. No unified, fair, and reproducible cross-schema benchmarking framework exists.

Research Gaps & Research Questions

Three Unresolved Research Gaps:

- 1 **(G1) Weak Dataset Auditability:**
Most studies cite “public datasets” without provenance, deduplication rules, or rebuild scripts.
- 2 **(G2) Unfair Parametric Baselines:**
COCOMO II applied with default parameters when cost drivers are missing $\Rightarrow$ *straw-man comparisons*.
- 3 **(G3) Ambiguous Aggregated Reporting:**
Micro-averaging dominated by LOC (90.5%), masking true FP/UCP behavior.

Research Questions

RQ1: Do ensemble methods (RF, XGBoost) outperform a *calibrated* baseline across LOC/FP/UCP?

RQ2: Does macro-averaging with LOSO provide a more honest evaluation than micro-averaging?

Primary Contribution

A **reproducible cross-schema benchmarking framework** for SEE — a reusable methodological artifact for the community.

Dataset: Auditable Manifest — 18 Sources (1979–2022)

3,054 Projects from 18 Sources:

Schema	Key Sources	Projects	Years
LOC	ISBSG, NASA93, China, ...	2,765	1971–2022
FP	ISBSG-FP, Maxwell, Albrecht	158	1974–2018
UCP	Ochodek, Diev, Karner	131	1993–2017
Total	18 sources	3,054	

Data Quality Controls (G1):

- Provenance documented with DOI/URL
- Cross-source deduplication rules
- Rebuild scripts for full reproducibility
- No test-set leakage

Preprocessing Pipeline (4 steps):

- 1 **Unit harmonization:** Effort → PM; LOC → KLOC
- 2 **Missing values:** Median imputation (features only)
- 3 **Outlier mitigation:** IQR-based capping
- 4 **Distribution shaping:** $\log_{1p}$ transform

Contribution #1

Full dataset manifest with rebuild scripts — addresses G1.

Multi-Schema ML Framework & Evaluation Protocols

Architecture: Schema-Independent Training

$$\hat{E}^{(s)} = f^{(s)}(x^{(s)}), \quad s \in \{\text{LOC}, \text{FP}, \text{UCP}\}$$

Five Models Evaluated:

- 1 Linear Regression (log-log variant)
- 2 Decision Tree Regressor
- 3 **Random Forest** (bagging, 500 trees)
- 4 **Gradient Boosting** (300 estimators)
- 5 **XGBoost** (regularized boosting)

Schema-Specific Protocols (G3)

LOC ($n=2,765$):

- 80/20 $\times$ 10 seeds + **LOSO** (11-fold)

FP ($n=158$, small sample):

- **LOOCV** (158-fold) + Bootstrap CI

UCP ($n=131$):

- 80/20 $\times$ 10 seeds + 5-fold CV

Contribution #3

Schema-appropriate evaluation protocols.

Two Core Methodological Innovations

Innovation #1 — Calibrated Baseline

(addressing G2)

Instead of COCOMO II with default parameters, fit a power-law model *on training data*:

$$\log(E) = \alpha + \beta \cdot \log(\text{Size}) \quad (1)$$

(α, β) via `curve_fit` (Levenberg-Marquardt), per-schema, per-seed.

⇒ **Fair lower bound**: ML must beat an already-optimized parametric model.

Innovation #3 — Tail Evaluation

Separate analysis of top 10% highest-effort projects.

Innovation #2 — Macro-Averaging

(addressing G3)

Problem: Micro-averaging

LOC = 90.5% ⇒ overall metric dominated by LOC.

Solution: Macro-averaging

$$m_{\text{macro}} = \frac{1}{3}(m_{\text{LOC}} + m_{\text{FP}} + m_{\text{UCP}}) \quad (2)$$

Each schema contributes **33.3%** regardless of n .

Combined Effect

All three innovations together address G1–G3 and ensure fair, reproducible comparison.

Overall Results: Macro-Averaged Across 10 Seeds

Model	MAE↓	MMRE↓	PRED(25)↑	R^2 ↑
Rand. Forest	12.66$\pm$0.85	0.647$\pm$0.041	0.395	0.812
XGBoost	13.21 $\pm$ 0.92	0.689	0.372	0.798
Grad. Boost	13.45 $\pm$ 0.98	0.701	0.361	0.789
Decision Tree	15.12 $\pm$ 1.20	0.812	0.318	0.734
Lin. Regr.	15.78 $\pm$ 1.15	0.845	0.302	0.712
<i>Calib. Baseline</i>	<i>18.45$\pm$1.20</i>	<i>1.12$\pm$0.08</i>	<i>0.254</i>	<i>0.621</i>

MAE: person-months; ↓=lower better; ↑=higher better

Answer to RQ1

RF vs. Baseline:

- MAE: 18.45 $\rightarrow$ 12.66 PM ($\downarrow$ 31%)
- MMRE: 1.12 $\rightarrow$ 0.647 ($\downarrow$ 42%)
- PRED(25): 0.254 $\rightarrow$ 0.395 ($\uparrow$ 56%)

Statistical Significance

Wilcoxon: $p < 0.001$

$t=12.34$, $p<0.001$; $d=1.23$

Per-Schema Results & Cross-Source Generalization

Schema	Model	MAE (PM)	MMRE	Improvement
LOC ($n=2,765$)	Random Forest	11.2	0.589	↓40%
	Calib. Baseline	16.8	1.024	
FP ($n=158$) [†]	Random Forest	15.8	0.731	↓38%
	Calib. Baseline	21.4	1.178	
UCP ($n=131$)	Random Forest	11.0	0.621	↓49%
	Calib. Baseline	17.2	1.216	

[†]FP results are exploratory (small n ; LOOCV protocol).

LOSO Generalization (RQ2)

LOSO MAE degrades only **21%** vs. random split
⇒ adequate cross-source generalization.

Tail Evaluation (top 10% effort)

RF: **57–72%** MAE degradation

Baseline: **83–100%**

⇒ Ensemble **superior** on long-tail.

Ablation Study & Threats to Validity

Ablation: preprocessing contribution (RF, LOC)

Configuration	MAE	MMRE
Full pipeline	11.2	0.589
w/o $\log_{1p}$ transform	13.4	0.712
w/o IQR capping	14.1	0.798
w/o unit harmonization	17.3	1.045
Raw (no preprocessing)	19.8	1.234

⇒ Unit harmonization is the most critical step.

Threats to Validity:

- **FP** ($n=158$): exploratory only
- **Units**: 1 PM = 160 hrs (org-dependent)
- **Transfer**: no cross-schema sharing
- **Scope**: 1979–2022 datasets

Validity Threats

Construct: MMRE bias (underestimation)

External: N. America/EU skew

Internal: Residual duplication risk

Conclusion & Future Work

Five Scientific Contributions:

- 1 **Dataset manifest** + rebuild scripts (18 sources, 3,054 projects)
- 2 **Calibrated baseline** — no straw-man comparisons
- 3 **Schema protocols** — LOSO/LOOCV/bootstrap CI
- 4 **Macro-averaging** — prevents LOC dominance
- 5 **Tail evaluation** + LOSO generalization testing

Key Results

RF: **42% MMRE improvement** ($p < 0.001$);
LOSO: only **21% MAE degradation**.

Future Directions:

- Cross-schema transfer learning
- LSTM/Transformer for time-series
- Story points / DevOps metrics
- Active learning for small- n FP/UCP

Core Message

The **primary contribution is the benchmarking framework itself** — reproducible, schema-specific, imbalance-aware; adoptable for future SEE comparisons.

Thank You!

We welcome questions from the committee.

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Note on Demo

This presentation does not include a live demo. Full source code, dataset rebuild scripts, and experimental logs are available upon request by the evaluation committee.

[Backup] Complete Dataset Sources (18 Sources)

Table: Dataset Manifest — 18 publicly available sources

Source	Schema	Projects	Years	Domain
ISBSG R2020	LOC	1,245	1997–2020	Multi-domain
NASA93	LOC	93	1971–1987	Aerospace
Promise Cocomonasa	LOC	60	1985–1987	NASA
Desharnais	LOC	81	1989–1991	Canada
COCOMO81	LOC	63	1964–1979	Embedded
Kemerer	LOC	15	1980–1984	Business
Kitchenham	LOC	145	1990–1995	Commercial
Albrecht	LOC	24	1974–1979	IBM
Maxwell	LOC	62	1993–1999	Finland
Miyazaki94	LOC	48	1977–1991	COBOL
China	LOC	929	1996–2022	Chinese
ISBSG FP subset	FP	67	1997–2018	Multi-domain
Maxwell FP	FP	41	1993–1999	Finland
Kemerer FP	FP	15	1980–1984	Business
Albrecht FP	FP	35	1974–1979	IBM
Karner	UCP	10	1993	OO systems
Ochodek	UCP	71	2009–2013	Academic
Diev	UCP	50	2012–2017	Industrial

[Backup] Hyperparameter Configuration

Random Forest:

Parameter	Value
n_estimators	500
max_depth	20
min_samples_split	5
min_samples_leaf	2
max_features	sqrt
bootstrap	True
oob_score	True

Grid Search: 3-fold CV, 1,280 configurations, selected by lowest MAE.

XGBoost:

Parameter	Value
n_estimators	300
max_depth	8
learning_rate	0.05
subsample	0.8
colsample_bytree	0.8
gamma	0.1
reg_alpha	0.01
reg_lambda	1.0

Early stopping: 50 rounds patience, validation MAE.

Table: Comprehensive statistical validation results

Test	Statistic	p-value	Conclusion
Paired t -test (RF vs. Baseline)	$t = 12.34$	< 0.001	Significant
Wilcoxon signed-rank	$z = 9.87$	< 0.001	Significant
Friedman test (all models)	$\chi^2 = 45.6$	< 0.001	Models differ
Nemenyi post-hoc (RF vs. others)	—	< 0.05	RF significantly better
Cohen's d (RF vs. Baseline)	1.23	—	Large effect
Cliff's Delta	0.78	—	Large effect

Interpretation:

- Both parametric and non-parametric tests confirm RF significantly outperforms baseline
- Effect sizes $d > 0.8$ indicate *practical* significance, not just statistical
- Results are consistent across all 10 independent random seeds

① Why Random Forest over Deep Learning?

DL requires more data (FP: $n=158$ only). RF is interpretable and stable on small tabular datasets.

② How is overfitting controlled?

LOOCV (FP), LOSO CV, 10 seeds, OOB score, IQR capping.

③ Why report MMRE despite its bias?

Supplementary metric for literature comparison; **MAE is primary**. Standard SEE practice (Shepperd & Kadoda, 2001).

④ Is dataset deduplication sufficient?

Deduplication by project ID, effort range, and year. Residual risk noted in Threats to Validity.

⑤ Why not use full COCOMO II as baseline?

Most FP/UCP datasets lack cost drivers — default parameters create straw-man comparisons (gap G2).